DAW - WEB INTERFACE DESIGN

UNIT 1. GRAPHIC INTERFACE PLANNING

29/09/2026

Professor:
Begoña Rodríguez Ferreras

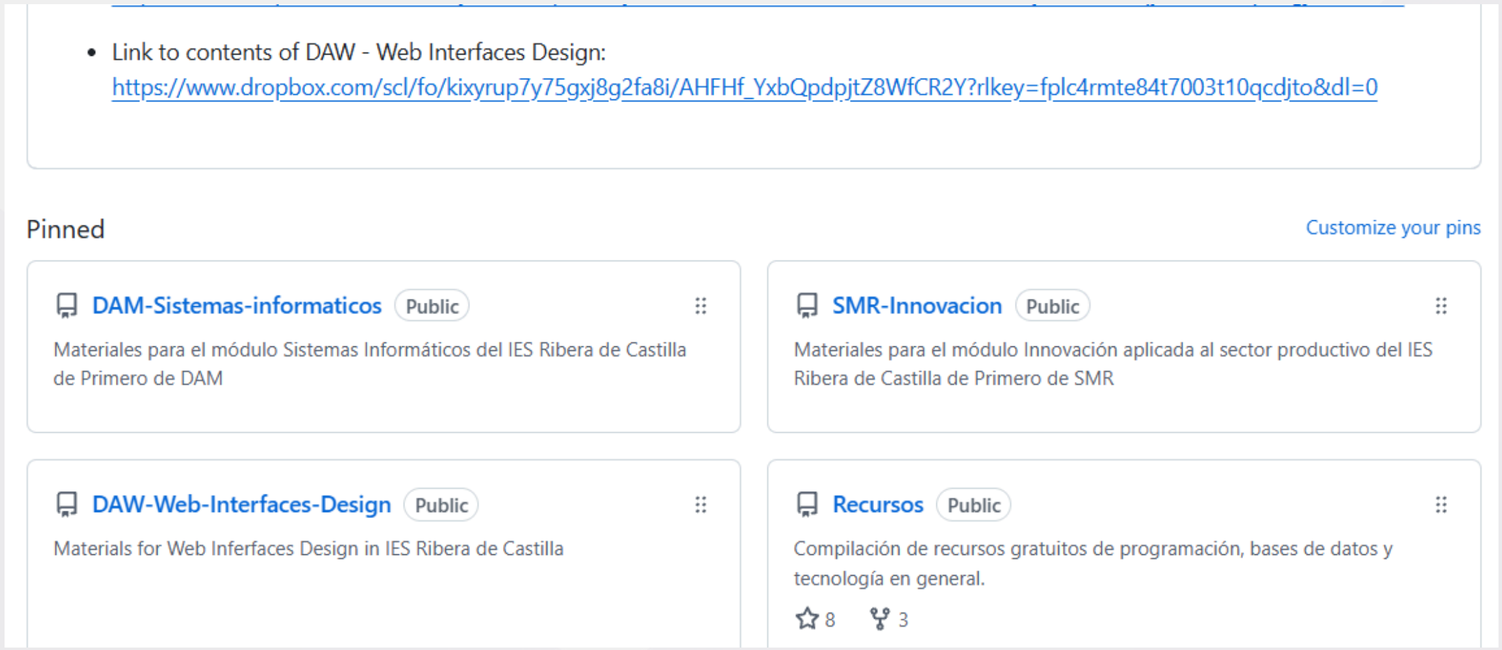
begona.rodfer.1@educa.jcyl.es

BOOK

INFORMÁTICA Y COMUNICACIONES
G.S. Desarrollo de Aplicaciones Web
Diseño de interfaces web
Diana García-Miguel López
2.ª edición
revisada y actualizada
EDITORIAL SÍNTESIS
Diseño de interfaces web
2.ª edición
revisada y actualizada

MATERIALS

GitHub: https://github.com/BegoRodriguez/DAW-Web-Interfaces-Design
Dropbox: https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0



TEAMS

I have created a classroom team for communication and project and activities delivery on Teams.

This is the code:

USER INTERFACE DESIGN FOR WEB DEVELOPMENT

Objectives:

- Recognize the importance of visual communication and its basic principles.
- Analyze different alternatives for presenting information in web documents.
- Identify and analyze the elements involved in creating web prototypes.
- Use and analyze content management systems as the foundation for developing a website.
- Plan and establish the process for designing a website.

DESIGN PRINCIPLES BASED ON HUMAN–COMPUTER INTERACTION

Human-Computer Interaction (HCI) is a field of study that focuses on the design and development of interactive systems that are easy and efficient to use.

HCI guidelines are recommendations that can be used to design and develop user interfaces that are more user-friendly and effective.

LICKLIDER & CLARK (1962)

Ten **problems to solve** for better Human–Computer Interaction (HCI):

- Sharing computer time among many users.
- An input–output system for communication using symbolic and graphical data.
- An interactive system capable of processing operations in real time.
- Systems for massive information storage that allow rapid retrieval.
- Systems that facilitate cooperation among people in the design and programming of large systems.

LICKLIDER & CLARK (1962) (II)

Ten **problems to solve** for better Human–Computer Interaction (HCI) (II):

- Recognition by computers of voice, printed handwriting, and direct handwritten data input.
- Automatic translation between natural languages.
- Automatic programming — enabling computers to generate or assist in writing programs.
- Self-organizing systems capable of adapting and improving their own performance.
- Logical and heuristic problem-solving systems that support human reasoning.

LICKLIDER & CLARK (1962) (III)

These ten problems anticipated almost everything modern computing tries to solve:

- multitasking
- graphical interfaces
- real-time systems
- databases
- collaborative development
- speech recognition
- machine translation
- AI-assisted coding
- adaptive systems
- expert systems

HANSEN (1972)

Wilfred J. Hansen's four principles for user-interface design, proposed in 1971, they are an important early contribution to Human–Computer Interaction (HCI).

Hansen presented them in “User Engineering Principles for Interactive Systems”, published in the proceedings of the 1971 Fall Joint Computer Conference.

The four principles are:

1. Know the user
2. Minimize memorization
3. Optimize operations
4. Engineer for errors

1. KNOW THE USER

The interface should be designed around the people who will actually use it, rather than around what is technically convenient for the programmer.

The designer should understand:

- the user's goals and needs;
- their previous experience;
- their knowledge and skills;
- the context in which they work;
- their limitations and likely difficulties.

2. MINIMIZE MEMORIZATION

Users should not have to remember unnecessary information in order to operate the system.

The interface should provide information that the user needs at the moment they need it.

For example, instead of requiring:
“*Remember that Ctrl+Alt+N creates a new network.*”
the system could provide a visible New network button or menu item.

This principle is closely related to a major modern HCI idea: recognition is generally easier than recall.

2. MINIMIZE MEMORIZATION (II)

You can see its influence in things we now take for granted:

- menus;
- visible buttons;
- autocomplete;
- contextual help;
- suggestions;
- recently used items;
- meaningful icons and labels.

3. OPTIMIZE OPERATIONS

The system should make users' actual tasks efficient.

This means reducing unnecessary actions, avoiding cumbersome procedures and designing operations around the user's workflow.

Design the system so that the user's work can be accomplished efficiently and naturally.

This is particularly relevant today for usability, keyboard shortcuts, automation, defaults and reducing the number of steps required to accomplish frequent tasks.

4. ENGINEER FOR ERRORS

Hansen recognized that users will make mistakes, and that a good system should therefore be designed to anticipate and handle them.

The important point is that errors should not simply be treated as the user's fault.
For example, if a user enters: 192.168.1.300
a good interface should detect the invalid address and explain the problem.

An even better interface might prevent invalid input from being entered in the first place.

4. ENGINEER FOR ERRORS (II)

This principle leads to several modern practices:

- validation.
- meaningful error messages.
- confirmation for dangerous operations.
- undo.
- recovery mechanisms.
- preventing invalid actions.
- preserving user data after an error.

CORE FUNCTIONS OF GRAPHIC DESIGN

Graphic design isn’t just “making things pretty.” In web development, it has functional, communicative, and cognitive purposes.

- **Communication & Information Delivery:** Translates raw data, copy, or abstract concepts into structured, easily digestible visual messages (think infographics, road signs, or annual reports).

- **Persuasion & Marketing:** Drives action and influences audience behavior in advertising, campaigns, and product packaging by appealing to specific emotions and needs.

- **Identity & Brand Building:** Establishes the distinct visual personality of a company, product, or organization through consistent logos, typography, color schemes, and imagery.

CORE FUNCTIONS OF GRAPHIC DESIGN (II)

Graphic design isn’t just “making things pretty.” In web development, it has functional, communicative, and cognitive purposes.

- **User Experience (UX) & Navigation:** Guides people through digital spaces (websites, apps) and physical environments (wayfinding in airports or subways) by creating intuitive visual hierarchies.

- **Engagement & Aesthetics:** Captures attention in a visual-heavy world, making content memorable and pleasant to interact with.

GROUPS OF ELEMENTS

- **Conceptual elements:** The ideas, concepts and meanings that the interface represents. Example: the concept of "deleting a file."

- **Visual elements:** What the user actually perceives: colours, typography, icons, shapes, images, etc. Example: a trash-bin icon representing deletion.

- **Relational elements:** The relationships between elements: hierarchy, proximity, alignment, grouping, sequence and navigation. Example: placing "Save" next to "Cancel", or grouping network settings together.

- **Practical elements:** The elements that allow the user to perform actions and accomplish tasks. Example: buttons, forms, sliders, menus and controls.

DEFINITION OF WEB INTERFACE

For all the above, we can conclude that:

*“A web interface is the set of **visual and interactive elements**, together with their **organization and layout**, that allows users to access information, navigate a website and interact with its content and functions efficiently.”*

If these elements are not carefully designed and coordinated, a website may still work technically, but achieving a good user experience will largely be a matter of chance.

EXAMPLES OF GOOD AND BAD DESIGN

THE 6 PHASES OF WEBSITE DESIGN

The web design process moves from abstract strategy to concrete implementation in six distinct stages.

01. STRATEGY	Discovery, goals, & audience definition
02. STRUCTURE	Sitemap, user flows, & wireframes
03. UI DESIGN	Visual style, design system, & mockups
04. BUILD	Front-end, back-end, & CMS integration
05. TEST	QA, performance, accessibility, & cross-browser
06. LAUNCH	Deployment & continuous iteration

PHASE 1: DISCOVERY & STRATEGY

- **Objective:** Define the project scope, business goals, and target audience.

- **Key Tasks:**

  - Conduct stakeholder interviews
  - **Analyze competitors**
  - **Create user personas**
  - Establish Key Performance Indicators (KPIs).

PHASE 2: ARCHITECTURE & WIREFRAMING (UX)

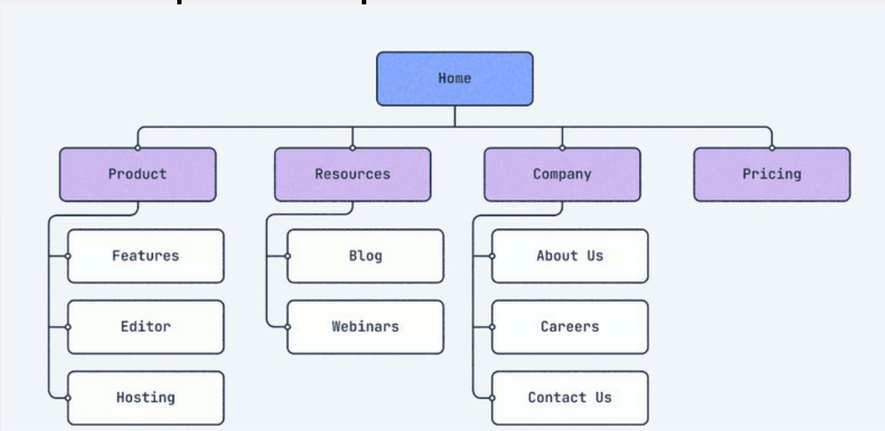
- **Objective:** Map out how content is organized and how users navigate the site.

- **Key Deliverables:**

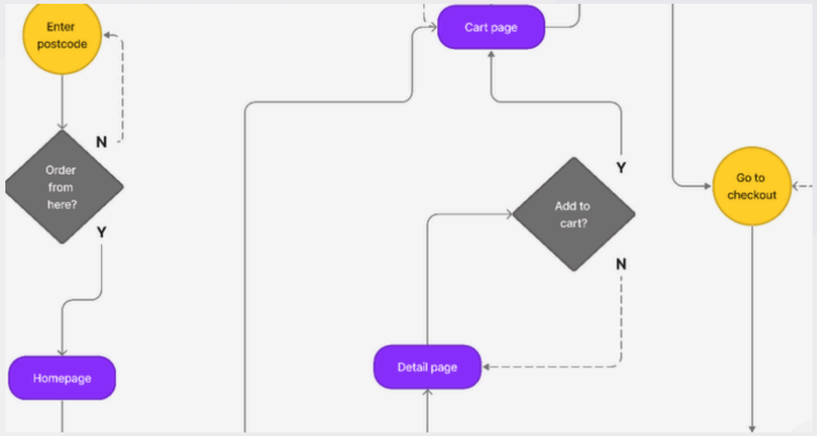
    - **Sitemap:** A structural tree of all pages.
    - **User Flows:** Step-by-step paths users take to complete key tasks (e.g., checkout, signup).
    - **Wireframes:** Low-fidelity grayscale blueprints focusing on element placement without visual styling.

PHASE 2: ARCHITECTURE & WIREFRAMING (UX) (II)

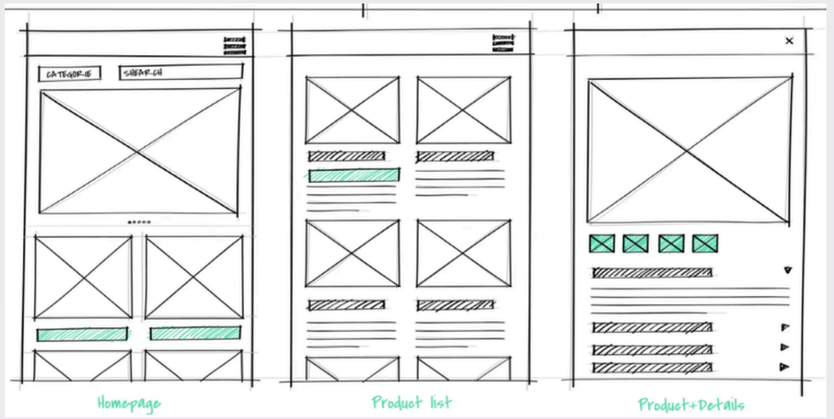
Sitemap example



User flow



Wireframe



PHASE 3: VISUAL DESIGN (UI)

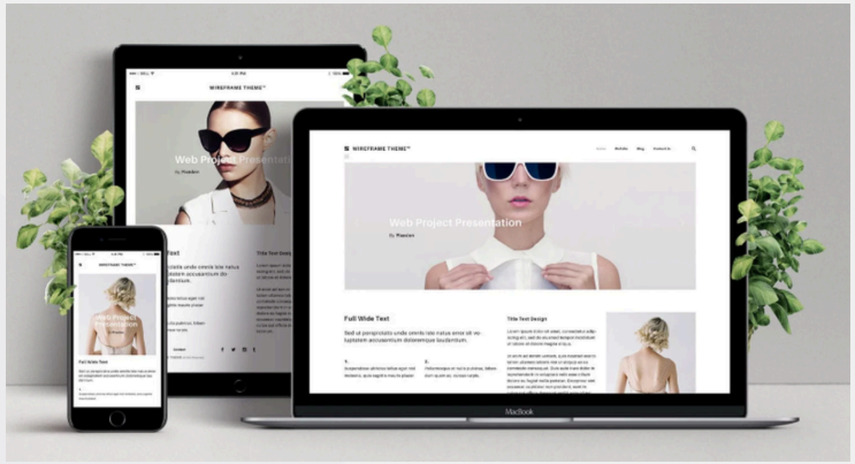
- **Objective:** Apply brand identity, color theory, and visual polish to wireframes.

- **Key Deliverables:**

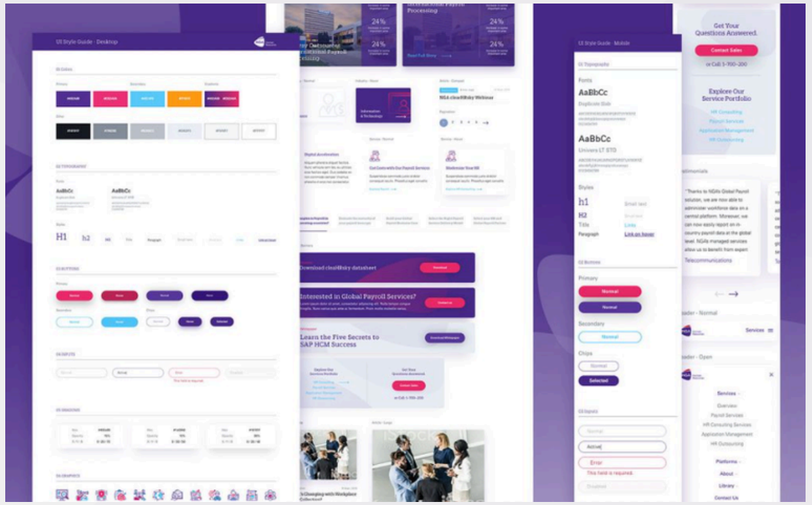
  - High-fidelity design **mockups.**
  - **Responsive layouts** (desktop, tablet, mobile).
  - **Interactive prototypes**.
  - A **design system** or **component library**.

PHASE 3: VISUAL DESIGN (UI) (II)

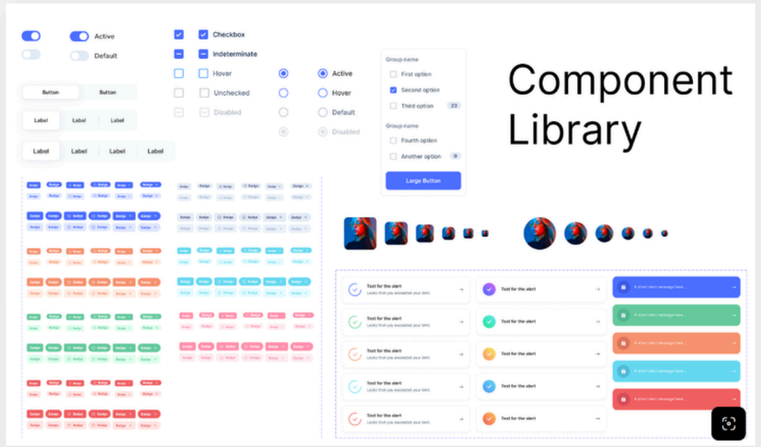
Responsive Layouts



Design system



Component library



PHASE 4: DEVELOPMENT

- **Objective:** Turn visual designs into functional code.
- **Key Tasks:**
  - Front-end development (HTML, CSS, JavaScript/frameworks)
  - Back-end/API setup
  - Database architecture
  - CMS integration (e.g., WordPress, Webflow, custom headless setup).

PHASE 5: TESTING & QUALITY ASSURANCE (QA)

- **Objective:** Ensure functional integrity, performance, and accessibility across all platforms.

- **Key Tasks:**

  - Cross-browser and cross-device responsiveness.
  - Web accessibility compliance (WCAG 2.1 AA).
  - Load time optimization and technical SEO best practices.
  - Form submissions and interactive feature validation.

PHASE 6: LAUNCH & OPTIMIZATION

- **Objective:** Deploy the site to production and monitor performance.
- **Key Tasks:**
  - Domain migration
  - SSL setup, analytics tracking setup (e.g., Google Analytics)
  - post-launch bug fixes
  - continuous conversion rate optimization (CRO).

TASK 1

Define the project scope, business goals, and target audience for your **Intermodular Project**.

- **Key Tasks:**
    - **Analyze competitors:** at least look for two webpages from direct competition in the area and identify the sitemap. We will analyze their design system later.
    - **Create user personas**
    - Establish Key Performance Indicators (KPIs): for example, client satisfaction.

APPLICATIONS: ANALYSIS, DESIGN, AND PROTOTYPES

Why mockups and prototypes matter

- They define the structure and layout early
- They accelerate development
- They help clients visualize the final product
- They reduce misunderstandings and rework

PROTOTYPES. KEY ELEMENTS

- A prototype is a mockup or model of a design that helps visualize the final product.
- Prototypes allow designers to explore different layout options, test basic functionality, and evaluate usability.
- They save time, effort, and money, since it is much easier to make changes to an early design than to modify the final product.

WHY PROTOTYPES MATTER IN WEB DESIGN

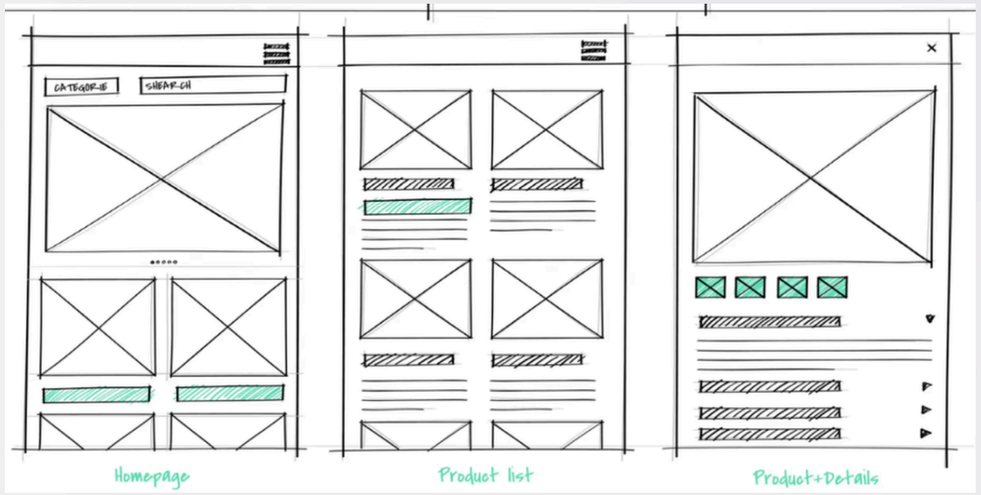
- They allow multiple design iterations
- They help refine menus and interface elements
- They involve the client early in the process
- They enable functionality and usability testing
- They reduce cost and rework before coding begins

TYPES OF PROTOTYPES: SKETCHING

- Paper-based drawings of the whole website
- Includes processes and screen relationships
- Used in the early design phase
- Defines content hierarchy
- No visual design details yet

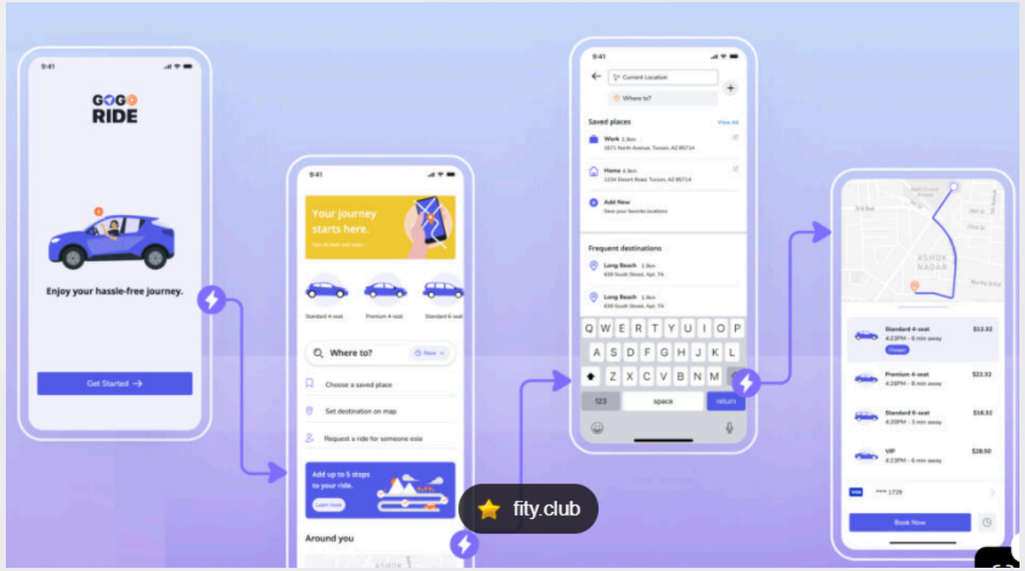
TYPES OF PROTOTYPES: WIREFRAMING

- More detailed screen drawings
- Includes content structure and **calls to action**
- Paper or digital format
- Produces a mock-up based on earlier sketches
- Helps validate design decisions with the client
- Main goal: content organization



TYPES OF PROTOTYPES: PROTOTYPING

- Digital design of screen interactions
- Final step before full development
- Evaluates design, content flow, and functionality
- Includes menus, forms, buttons, icons...
- Enables user testing
- Reduces development time and number of app versions



WIREFRAMING ORGANIZATION

The design of the prototype, in any of its phases, should be based mainly on the following aspects:

- Identifying the elements that belong to each page of the website.
- Distributing the number of graphical interface elements so that the screen is not overloaded, while still providing enough information and ensuring correct interaction.
- Organizing the hierarchy, order, and arrangement of elements.
- Ensuring an appropriate layout that efficiently uses the available space depending on the device.
- Selecting web design patterns to standardize the interface design.
- Evaluating technical aspects of usability and accessibility.

WIREFRAMING ORGANIZATION (II)

When discussing graphical user interfaces, two clearly differentiated areas can be identified. These are:

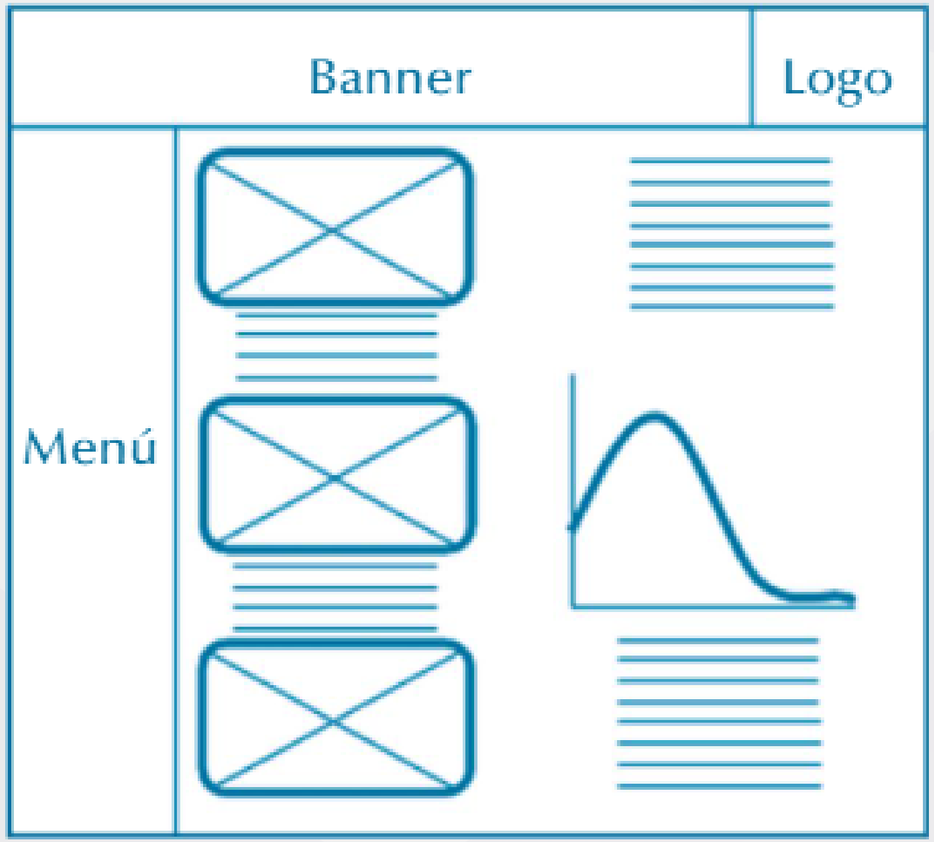
a) **Wireframes**, or specific content blocks, wich **varies from one page to another**.

b) **Generic layouts**, or **common parts** that are reused across multiple pages of the website (header, footer...).

WIREFRAMES: INITIAL STRUCTURE OF A WEB INTERFACE

- Begin by creating a schematic representation of the main parts of the interface.
- Build a wireframe, a low-fidelity visual prototype that acts as the skeleton of the website or web application.
- Use wireframes to define the initial layout and distribution of content.
- Avoid including final design elements; wireframes are created in black and white or grayscale.
- Focus on structure, hierarchy, and placement—not aesthetics.

WIREFRAMES:
INITIAL STRUCTURE OF A WEB INTERFACE (II)



ADVANTAGES OF WIREFRAMES

- One of the main advantages of wireframes is that their creation focuses solely on content architecture, not on visual design.

- Although prototypes are the tools that allow testing content interaction, some applications enable web page prototyping, meaning you can click through links and share early interactive versions with clients.

- These interactive wireframes help validate structure and navigation before building the final product, reducing development time and minimizing the number of required versions.

TASK 2A

Create a sketching and wireframing to design the user interface for the inter-module project (including mobile devices).

Use pencil (or pen) to quickly sketch the different screens and user flows, including the home screen.

Experiment with different layouts and design elements

WEB APPLICATIONS FOR CREATING WIREFRAMES

- **Figma** – Collaborative, cloud-based tool for creating wireframes, mockups, and interactive prototypes.
- **Adobe XD** – Intuitive interface design tool with strong prototyping features.
- **Balsamiq** – Low-fidelity wireframing tool that mimics sketch-style layouts for quick ideation.
- **Sketch** – Popular macOS design tool with plugins for wireframing and prototyping.
- **MockFlow** – Web-based platform for planning interfaces and creating structured wireframes.
- **Wireframe.cc** – Minimalistic tool focused on fast, distraction-free wireframe creation.

FIGMA: A COLLABORATIVE INTERFACE DESIGN PLATFORM

- Cloud-based tool for designing **wireframes, mockups and interactive prototypes**.
- Enables **real-time collaboration**, allowing multiple users to work on the same file simultaneously.
- Provides an intuitive interface with **vector editing**, reusable components, and design libraries.
- Includes built-in **prototyping features** to simulate navigation and user flows.
- Works directly in the browser—no installation required.
- Ideal for team projects, classroom activities, and professional UI/UX workflows.

TASK 2B:
DIGITAL PROTOTYPING IN FIGMA

- Based on the paper prototype created in Task 2a, now move to the digital design phase.
- Use Figma to recreate the interface, maintaining the structure and layout defined in your sketches.
- Translate each screen into a digital format, applying consistent spacing, alignment, and component organization.
- Ensure that user flows designed on paper are fully represented in the digital version.
- Prepare the prototype for future interaction testing and refinement.

CONTENT MANAGEMENT SYSTEMS (CMS)

- In addition to prototyping tools, there are applications dedicated to **content management**, known as **CMS (Content Management Systems)**.
- These programs allow users to **create, edit, and publish web content** through a graphical interface **without requiring programming knowledge**.
- CMS platforms are important because they provide an **attractive and consistent design**, adding value that can make users prefer one website over another.
- The choice of a CMS depends largely on the **purpose and requirements of the project**.

TYPES OF CMS

- CMS platforms can be classified into **two main categories**:
  - **Proprietary software** – Requires a valid license to use.
  - **Open-source software** – Allows developers to access and modify the source code, making customization easier.
- Most well-known CMS platforms are built using **web technologies** such as **PHP/HTML** and rely on database management systems like **MySQL**.

TEMPLATES IN CMS

- Building a website with a CMS relies on predefined design elements known as **templates**.
- Templates can be **fully customized** by the user from scratch.
- Alternatively, CMS platforms offer **pre-designed sites**, which provide a ready-made structure where users only need to add their specific content.
- These templates simplify development, ensure visual consistency, and speed up the creation of functional websites.

EXAMPLES OF CMS

- **WordPress** – The most widely used CMS; ideal for blogs, corporate sites, and small businesses.
- **Joomla** – Flexible and powerful, suitable for more complex websites and community platforms.
- **Drupal** – Highly customizable and secure; used for large-scale, enterprise-level projects.
- **Moodle** – A CMS specialized in e-learning; widely used in educational institutions.
- **Prestashop / Magento** – CMS platforms focused on e-commerce and online stores.
- **Ghost** – Modern, minimalist CMS designed for professional publishing and newsletters.

WORDPRESS

- WordPress is the world’s most widely used CMS, powering more than 40% of all websites on the internet.

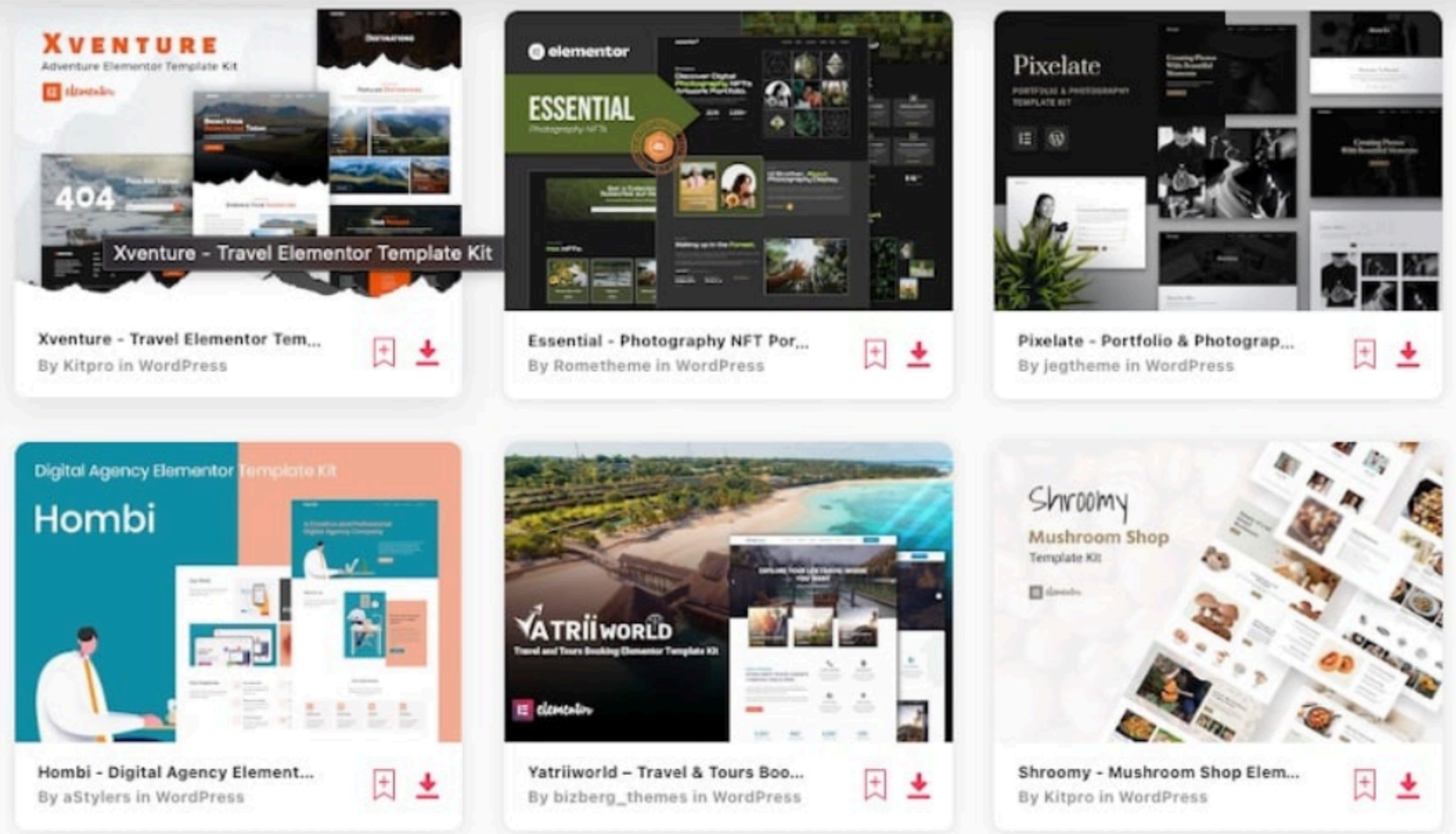
- Allows users to **create, edit and publish website** content through an intuitive dashboard—no advanced coding required.

- Offers thousands of **themes** to control the site’s appearance and **plugins** to extend functionality (SEO, security, e-commerce, forms, etc.).

- Available in two versions:
  - **WordPress.com** – hosted, easier for beginners.
  - **WordPress.org** – self-hosted, fully customizable.

- Suitable for blogs, business sites, portfolios, e-commerce stores, educational platforms, and more.

WORDPRESS THEMES



WEBSITE DESIGN: DEVELOPMENT PHASES

1. Definition of the elements that will form the application and website. Ensure there are enough elements, avoiding overload or saturation.

2. Determination of the **website's scope**, including its overall size and structure.

3. Selection of **web design patterns** to standardize interface design and ensure **consistency**.

4. Choice of technical aspects related to **usability** and **accessibility**, ensuring the site is easy to use and accessible to all users.

VISUAL HIERARCHY IN WEB DESIGN

- **Visual hierarchy** refers to the arrangement and ordering of elements on a page.
- It is not trivial: the **effectiveness and quality** of a website depend on defining clear **communication, information, and interaction priorities**.
- Elements should be placed from left to right and top to bottom to establish a natural reading flow.
- The **most important elements** are positioned **at the top-left**, and as elements move **downward and to the right**, they gradually lose emphasis and importance.

EDITORIAL AREA: DEFINING THE PROJECT FOUNDATIONS

- The **editorial area** establishes the fundamental pillars of the web project.
- Its main purpose is to define the **final objective** of the website.
- Different goals require different design approaches:
  - A website dedicated to **selling tire replacement parts** has different needs...
  - ...than a website focused on **selling concert tickets**.
- Clearly defining the website's purpose ensures coherent decisions in structure, content, and interface design.

PRODUCTION AREA: ALIGNING THE PROJECT WITH REAL-WORLD CONSTRAINTS

- Once the final purpose of the project has been defined, it must be aligned with **current market conditions**.
- This involves conducting a **feasibility study** and outlining the most appropriate **development plan**.
- Key agents involved in the production process include:
  - **Client:** commissions the creation and development of the website.
  - **User:** the target audience of the website; understanding their needs and expectations is essential to improve performance.
  - **Budget:** the financial amount the client allocates to the project; this influences the choice of CMS and other technical decisions.
  - **Work plan:** the schedule of deliverables and the distribution of tasks throughout the project.

TECHNICAL AREA: FROM REQUIREMENTS TO INTERACTIVE PROTOTYPES

- The **technical area** is formed by the technical specialists responsible for analyzing the project's programming requirements (typically HTML, CSS, and databases).
- Up to this point, the other areas have worked through **sketching** and later **wireframing**.
- The technical area begins developing the **first prototypes and mockups**, transforming conceptual designs into functional structures.
- These early prototypes start incorporating **initial interaction elements** (menus, buttons, forms), progressively evolving toward the **final product.**

ARTISTIC AREA: DEFINING THE VISUAL IDENTITY OF THE PROJECT

- The **artistic area** focuses on the final visual aesthetics of the project, ensuring they align with the website’s purpose, user reports, and client requirements.
- The **style** of the project will vary depending on the tools used and the characteristics of the target users.
- The visual style must become a **distinctive identity** for the website.
- The success of good graphic design lies in giving the site its **own personality**, making it **instantly recognizable** in any context.

USER INTERFACE DESIGN FOR WEB DEVELOPMENT

Objectives:

- Recognize the importance of visual communication and its basic principles.
- Analyze different alternatives for presenting information in web documents.
- Identify and analyze the elements involved in creating web prototypes.
- Use and analyze content management systems as the foundation for developing a website.
- Plan and establish the process for designing a website.

WEB DESIGN ELEMENTS: COLOR, HUE, AND SATURATION

Objectives:

- Recognize the importance of visual communication and its basic elements.
- Analyze the role of color in website design.
- Identify and create a web mockup representing the main parts of a webpage.
- Differentiate the components of a web interface.
- Understand and apply basic design elements in any web document.

WEB INTERFACE: NAVIGATION AND CONTENT

One of the most important aspects in web interface development is **page layout**, meaning the definition of the main elements that structure a webpage and the areas they occupy.

We will have two main areas:

1. Navigation Area

2. Content and Interaction Area

1. NAVIGATION AREA

- This area provides access to all the content of the website.
- Navigation must give users a clear sense of where they are and how they can move through the site.
- To achieve this, the site should include at least two elements:
  - An indicator showing the **current location** within the site.
  - A global content index that can be consulted at any time.

2. CONTENT AND INTERACTION AREA

- This is the main area of a website, where content is displayed.

- It includes text sections, input fields (with labels indicating what information should be entered), images, videos, and any other elements that make up the site.

WEB LAYOUT: WHAT DOES “LAYOUT” MEAN IN WEB DESIGN?

- In web design, **layout** refers to creating a **preliminary prototype** that defines how the interface will organize and present all required elements.

- As in any design process, having a **prior structure** (prototype) is essential to ensure the interface meets all necessary requirements.

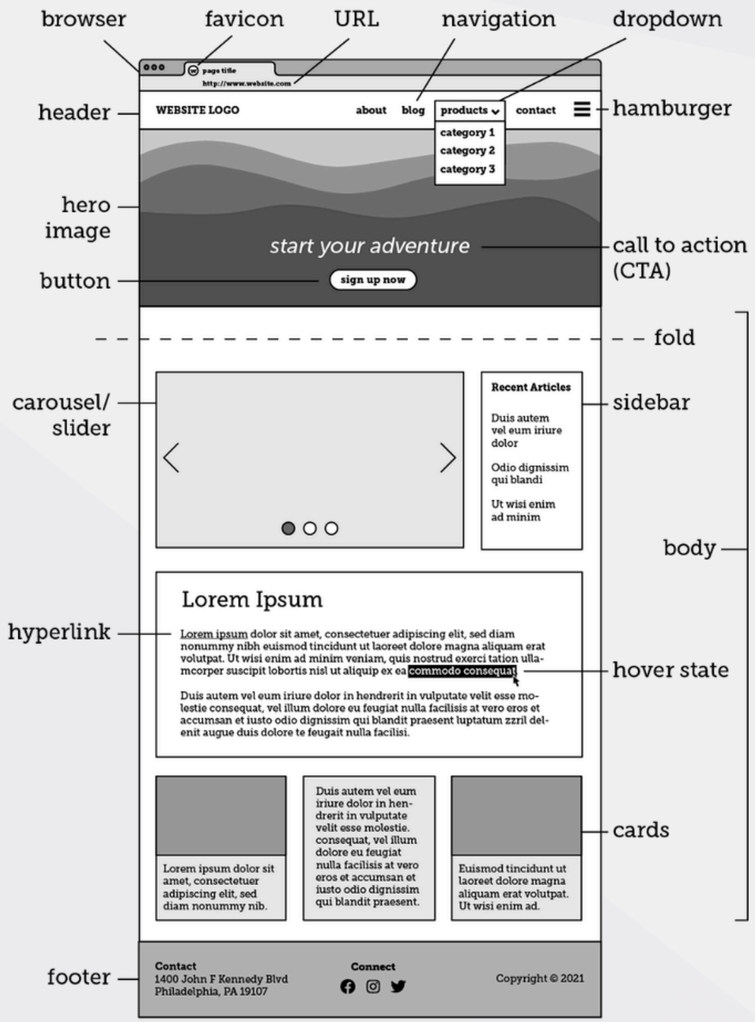
KEY QUESTIONS FOR STARTING A WEB PROTOTYPE

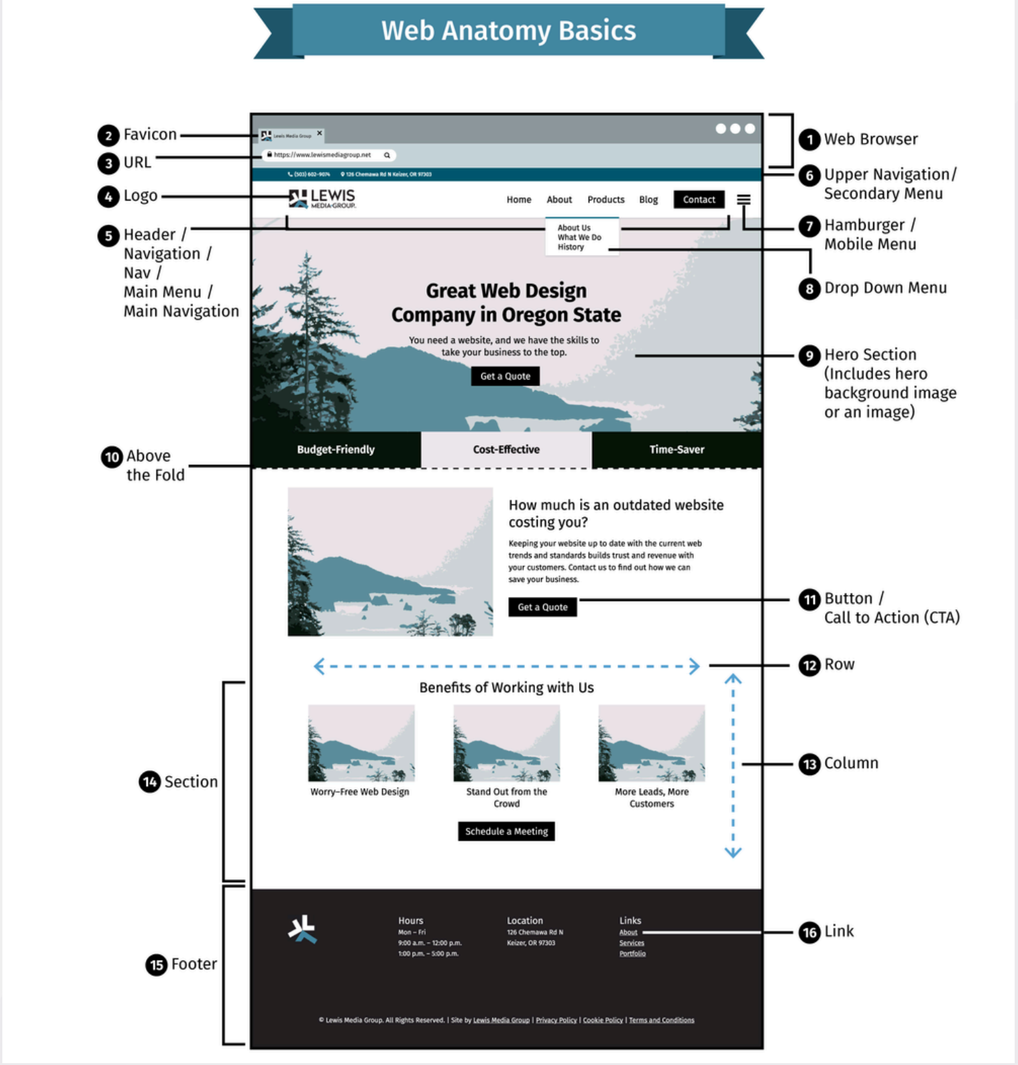
1. What elements will the website include?

2. How many pages will the site contain?

3. How will the elements be distributed across each page?

ELEMENTS OF A WEBSITE





COMMON ELEMENTS OF A WEB PAGE

- Most websites share a set of common structural elements, though none of them are strictly mandatory.

- A designer may choose to omit elements (such as the footer), but including them is generally recommended for clarity and usability.

- Common elements typically include:
  - Header: logo, navigation
  - Navigation menu
  - Body: Hero section, Main content area, Sidebar (optional)
  - Footer: recommended, but not required

HEADER

- The header is responsible for the **visual identification** of the website, giving it a recognizable image or brand.

- It is common for the header to appear on **all pages** of the site.
- Although the typical header is a **rectangular bar at the top**, modern designs often use **larger or more expressive layouts** that provide a broader visual overview of the site.
- In some cases, the header may not appear on every page, but it **must always be present on the homepage**.

NAVIGATION SYSTEMS

- The **navigation system** displays the names of all the pages that make up the website.
- It is usually placed **right below the header**, although it is also common to find it on the **left side** of the page.
- By clicking on each page name, users can access the desired section of the site.
- Effective navigation helps users understand **where they are** and **how to move** through the website.

WHAT A ROLLOVER IS

- A **rollover** is a visual change triggered when the mouse pointer enters or leaves an element. This can include:
    - Color changes
    - Image swaps
    - Shadows or highlights
    - Size or border changes
    - Small animations
- These effects provide immediate visual feedback, helping users understand that an element is interactive.

FOOTER

- The footer is located at the bottom of a webpage, positioned below the main content area.

- It usually contains important information about the site, such as:

  - Credits
  - Licensing information
  - Links to contact forms
  - Additional legal or structural information (privacy policy, terms of use, etc.)

- Although a footer is not mandatory, including one is generally recommended for clarity, credibility, and user support.

BODY

- The **body** is the **central part** of a webpage, where the information intended for the user is displayed.
- It is positioned **below the header** and the **navigation menu**.
- All elements placed in the body should maintain a **consistent and uniform design** across the different pages of the website.
- A coherent body layout improves **usability, visual harmony,** and the **overall user experience**.

HERO SECTION

- A hero section is the large introductory block at the top of a webpage that typically includes:
  - A **clear headline** explaining what the site or product is about.
  - A **subheadline** that adds context or value.
  - A **strong visual** (image, illustration, or video) that represents the brand or product.
  - A **primary call-to-action (CTA)** guiding the user on what to do next.

NAVIGATION MAP

- Websites often have a complex structure of pages and sections.
- Some pages can be accessed from anywhere (like the homepage), while others are only reachable from specific locations within the site.
- For this reason, during the layout and planning phase, it is advisable to create a diagram (**navigation map**) showing all sections/pages and the relationships between them.
- Navigation maps provide a schematic representation of the website’s structure, showing its pages, sections, and how they connect. It helps ensure an optimal site design.
- They act as a kind of content index, allowing users to locate what they need more quickly.

NAVIGATION DESIGN

Jakob Nielsen (2000), in “Designing Web Usability: The Practice of Simplicity” proposes three fundamental questions to simplify and improve navigation system design:

1. Where am I?

- Users must always know their exact location within the website.
- If they cannot find what they need easily, they may abandon the site.
- The page should be defined in relation to the overall site structure.
- A common solution is a **navigation map** or **menu** that highlights the current page.

NAVIGATION DESIGN (II)

2. Where have I been?

- Users should be able to identify which pages they have already visited.
- This is often implemented by changing the color of visited links.

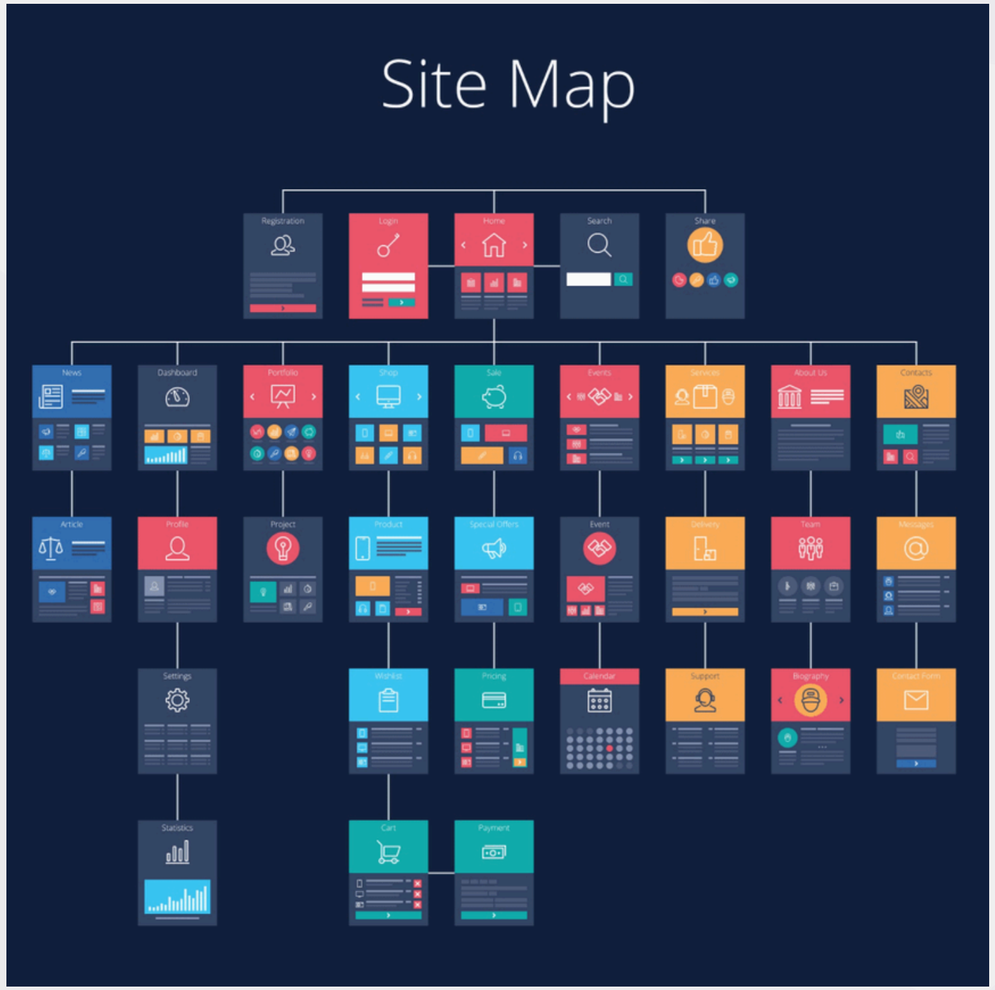
3. Where can I go?

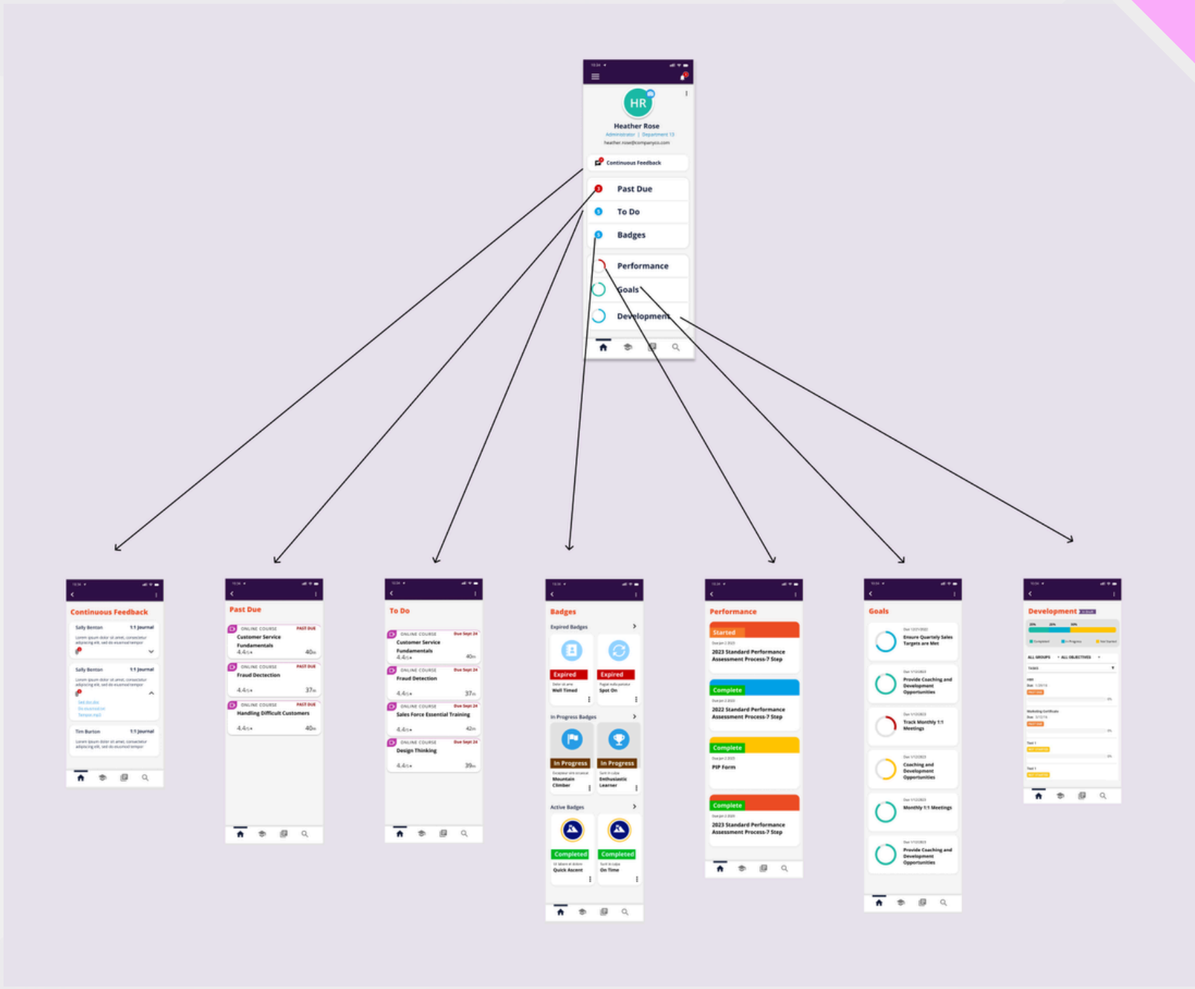
- Navigation systems should clearly show all available paths for moving through the site.
- Recommended to display all necessary links that allow users to navigate efficiently.

NAVIGATION MAP (II)

- When representing a navigation map, its structure often resembles a concept map (also a sitemap).
- Both include a root node, which in navigation maps corresponds to the homepage.
- From this root node, all other elements (pages and sections) of the site branch out.
- “Child” elements may also be interconnected, allowing users to move between pages without returning to the homepage.
- This structure helps visualize how users can navigate through the site and how different pages relate to one another.

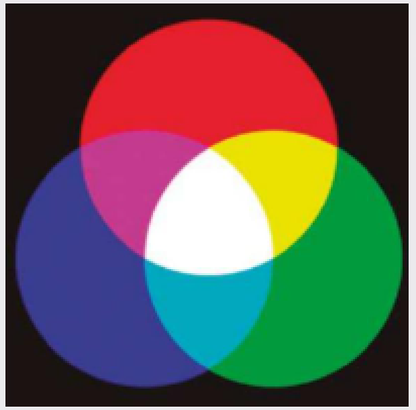
NAVIGATION MAP (III)





COLOR: THE RGB SYSTEM

- The human eye perceives color through three types of cone cells, each sensitive to wavelengths corresponding roughly to red, green, and blue.
- Digital devices use the same principle: the RGB (Red–Green–Blue) additive color model combines different intensities of these three light channels to produce all visible colors.
- By adjusting the proportion of red, green, and blue, a computer can generate the entire known color palette on screens.
- Because RGB is additive, adding all three colors at full intensity produces white, while zero intensity produces black.



RGB COLOR ENCODING

- Each RGB channel (Red, Green, Blue) is encoded using 8 bits, meaning each channel can represent 256 distinct values (because 2^8 =256).

- These values range from 0 to 255, where 0 means no intensity and 255 means maximum intensity.

- Combining the three channels (R, G, B) produces more than 16.7 million colors in standard 24-bit color (8 bits × 3 channels).

- Colors can be represented in decimal RGB (e.g., rgb(51, 102, 204)) or in hexadecimal (e.g., #3366CC). Both notations encode the same values: each HEX pair corresponds to one RGB channel.

HUE, SATURATION, AND BRIGHTNESS

- Beyond the RGB values assigned to each color channel, colors have three key properties that help distinguish one from another: hue, saturation, and brightness.
- **Hue** refers to the specific color family (red, blue, green, etc.).
- **Saturation** indicates how intense or pure a color is; low saturation produces more muted or grayish tones.
- **Brightness** (or value/lightness) describes how light or dark a color appears.
- These properties allow designers to classify colors as chromatic, complementary, or analogous, and to define color contrast effectively in visual compositions.

HUE

- Hue is the attribute that allows us to distinguish one color from another.
- The three primary hues in the additive RGB system are red, green, and blue.
- All other hues are produced by mixing these three primary colors in different proportions.
- Hue also helps define complementary colors, which are pairs of colors positioned opposite each other on the color wheel.
- Understanding hue is essential for creating effective contrast and harmonious color relationships in design.

SATURATION

- Saturation defines the intensity or purity of a color.
- It can be related to the bandwidth of light being perceived, and therefore depends on the amount of gray present in the color.
- The more gray a color contains, the lower its saturation becomes, resulting in a more muted, less intense appearance.
- Highly saturated colors appear vivid and pure, while low-saturation colors look soft, pale, or grayish.

BRIGHTNESS

- Brightness is the attribute that defines the amount of light a color contains.
- It represents how dark (when black is added) or light (when white is added) a color appears relative to its pure, unmodified version.
- Increasing brightness makes a color look lighter and closer, while decreasing brightness makes it appear darker and farther away in a graphic composition.
- In design, brightness is essential for creating depth, visual hierarchy, and contrast.

COLOR SELECTION IN INTERFACE DESIGN

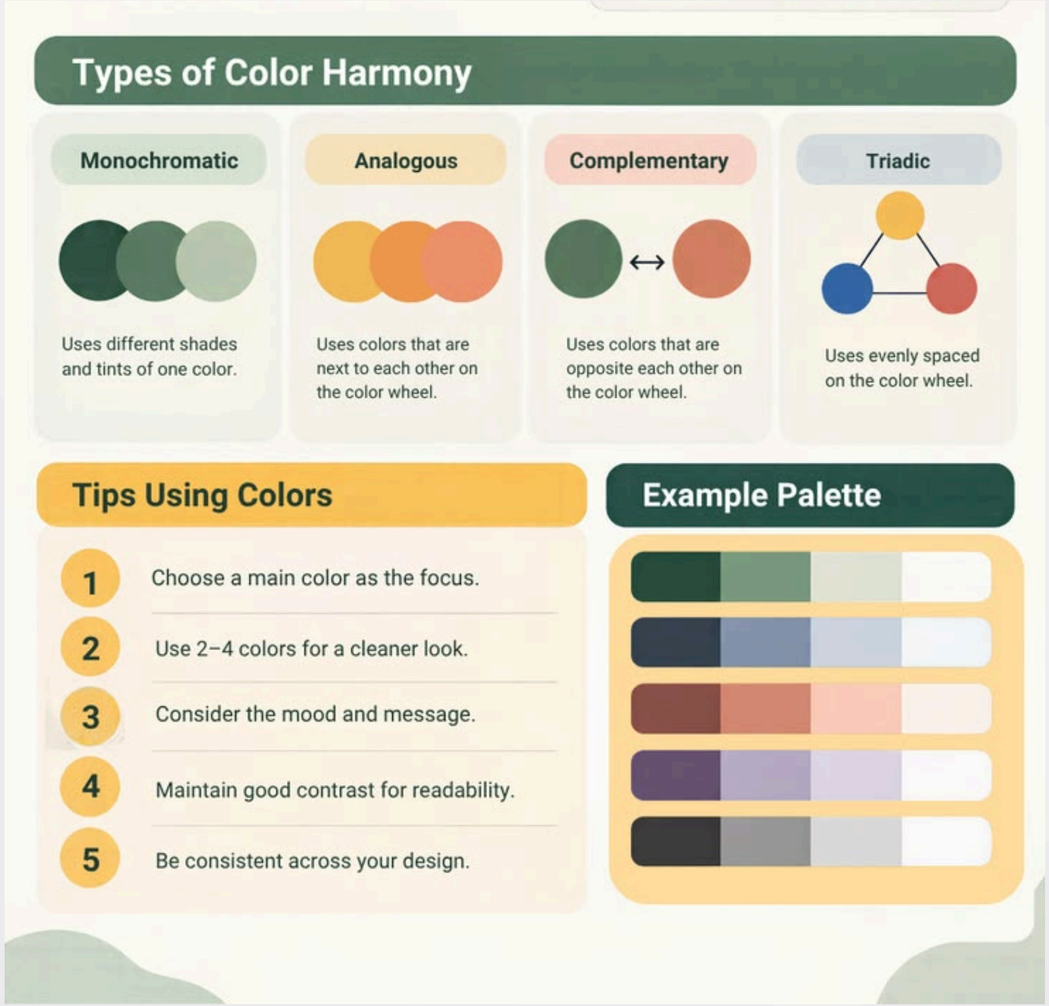
- Choosing an appropriate color palette is essential in graphical interface design, as it can strongly influence — even determine — the user’s navigation experience.

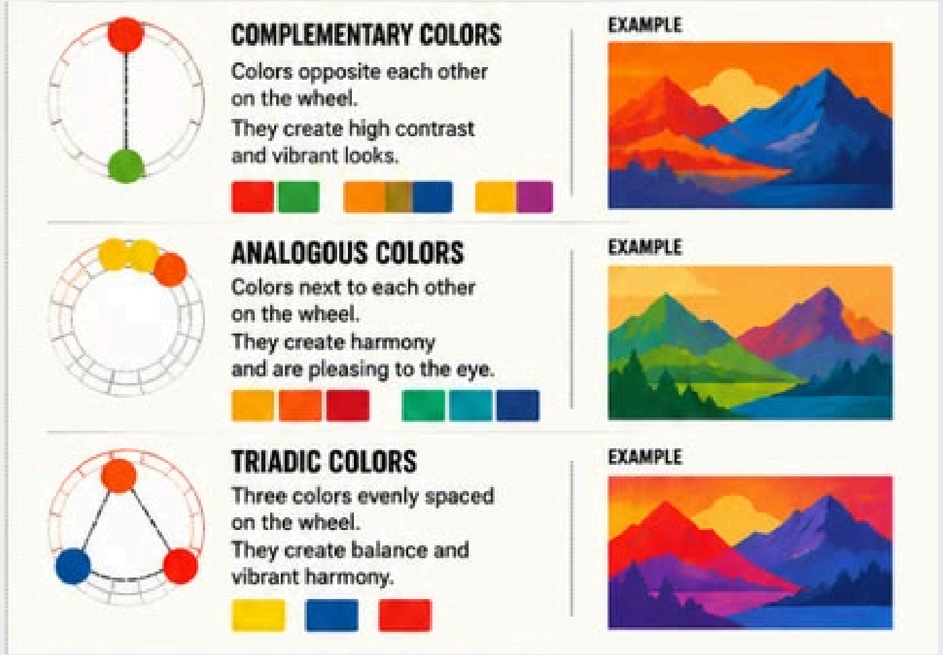
- The simplest option is **monochromy**, which consists of selecting a single hue from the color wheel and creating variations by adding white or black to generate different tones.

- Another option is using neighboring (analogous) colors, which are harmonious and provide visual stability. **Neighboring colors** are those located within a 90-degree range on the color wheel.

- These strategies help create consistent, readable, and aesthetically balanced interfaces.

COLOR SELECTION IN INTERFACE DESIGN (II)





COLOR THEORY

Smart Mentors
Premium Web Education Academy

Understanding Color Psychology

Color Theory in UI Design

WEB-SAFE COLORS

- Web-safe colors are colors that appear exactly the same across all browsers, devices, and operating systems.
- Their purpose is to ensure a consistent color experience for all users, regardless of hardware or software differences.
- Although modern devices can display millions of colors, the web-safe palette remains a reference for maximum compatibility.
- Web-safe colors are created exclusively by combining the following hexadecimal values for each RGB channel: 00, 33, 66, 99, AA, CC, FF
- These values produce a palette of 216 colors that were historically guaranteed to render uniformly.

WEB-SAFE COLORS (II)

TRUE COLOR

- True color refers to any digital system capable of representing color using 24 bits per pixel, divided into 8 bits per channel for red, green, and blue.
- Each channel can express 256 intensity levels (0–255), producing a total of 256×256×256=16,777,216 possible colors — commonly known as 16.7 million colors.
- This number of colors exceeds the approximate 10 million colors the human eye can distinguish, making 24-bit depth sufficient for lifelike color representation.
- In true color systems, each pixel is represented by 3 bytes (one per RGB channel).
- Modern screens, cameras, and most digital formats (PNG, JPEG, WebP) use at least 24-bit true color, ensuring smooth gradients and natural-looking images.

DESIGN ELEMENTS

1. Conceptual Elements (Point, Line, Plane, Volume): These define the *basic conceptual building blocks* of any visual composition.

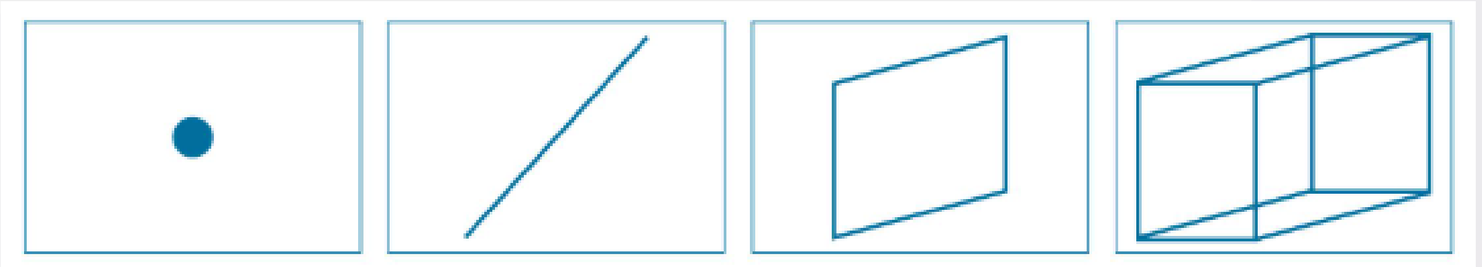
2. Visual Elements (Shape, Size, Color, Texture): These describe what the viewer *actually sees*.

3. Relationship Elements (Direction, Position, Space, Gravity): These define how elements *relate to each other* within a composition.

4. Practical Elements (Representation, Meaning, Function): These determine how a design *communicates, what it conveys* and *how it works*.

CONCEPTUAL ELEMENTS

- The basic conceptual elements are: point, line, plane, and volume.
- Conceptual elements are not visible; they have no physical existence.
- They interrelate with one another, generating more complex and meaningful figures.



THE POINT

1. Indicates position
2. Has no length
3. Has no width
4. Does not occupy space
5. Is the beginning and end of a line, and the spot where two lines intersect
6. A sequence of points becomes a line

THE LINE

1. Has length
2. Has no width
3. Is defined by points
4. Forms the edges of a plane
5. The movement or path of a line becomes a plane

THE LINE

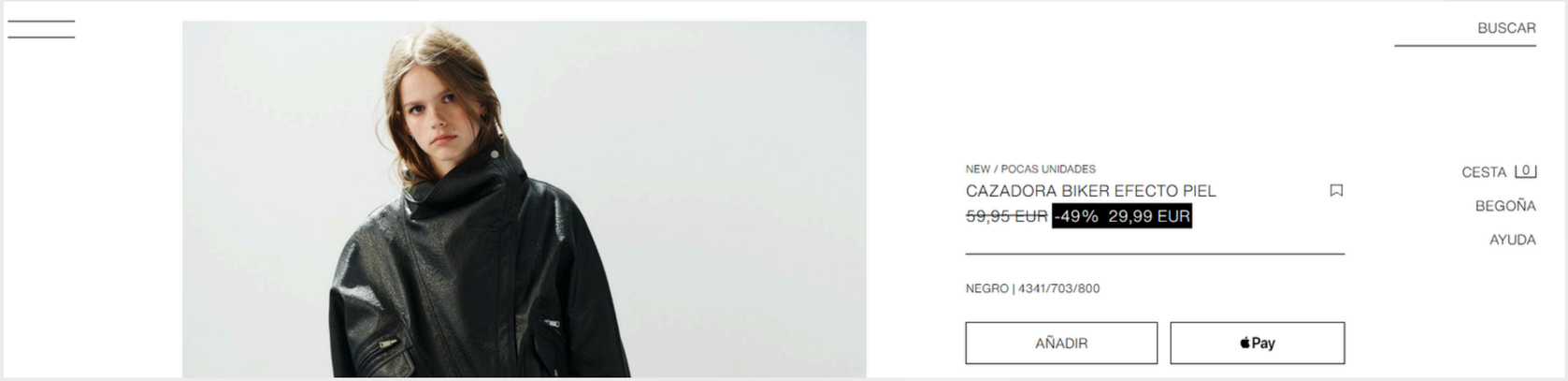
A line is any mark connecting two points. That sounds simple, but lines carry enormous expressive weight.

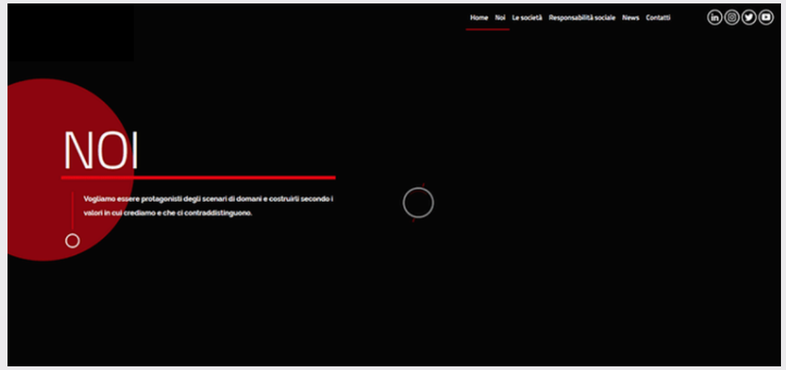
- **Horizontal** lines feel **calm** and **stable** — that's why landscape photographs feel serene.
- **Vertical** lines feel **strong** and **tall** — that's why skyscraper photography feels powerful.
- **Diagonal** lines create **tension** and **movement** — that's why action movie posters use them constantly.

LINE: PRACTICAL APPLICATION

- Use **thin, light lines** to separate content sections without adding visual weight.
- Use **thick, bold lines** to draw attention or create emphasis.

A single well-placed line can do the work of an entire decorative element.
In web design, a 1px horizontal rule between sections is often more effective than a gradient divider or decorative graphic.





THE PLANE

1. Has length
2. Has width
3. Has no thickness
4. Has position and direction
5. Is defined by lines
6. Marks the outer limits of a volume

THE VOLUME

1. Has a position in space

2. Is defined by planes

3. In two-dimensional design, volume is illusory

VISUAL ELEMENTS

- Previously we reviewed the basic conceptual elements, which allow the creation of more complex shapes and structures.

- These elements also have visual characteristics that define them: shape, size, color, and texture.

SHAPE

- Every element has shape as its main characteristic.
- This attribute identifies the object it represents.
- A shape is created when a line touches another line or itself, enclosing a plane — an area perceived as distinct from the surrounding space.
- There are two types of shapes:
  - Two-dimensional, called figures
  - Three-dimensional, perceived as forms

MEASURE

- Every element and its shape have a size

- This attribute is relative, meaning it is defined in relation to the dimensions of other shapes

COLOR

- An element — and the shape that forms it — always has a color that distinguishes it from its surroundings
- When several identical shapes must be differentiated, color is the attribute used
- Any color can be used, in any of its tonal or chromatic variations
- However, as previously noted, the final color will depend on the characteristics of the system being used

TEXTURE

- Just as a shape acquires measure and color that distinguish it from others, it also has the visual attribute of texture

- Texture refers to the appearance of a shape’s surface, which may be flat or embossed, smooth or rough, ...

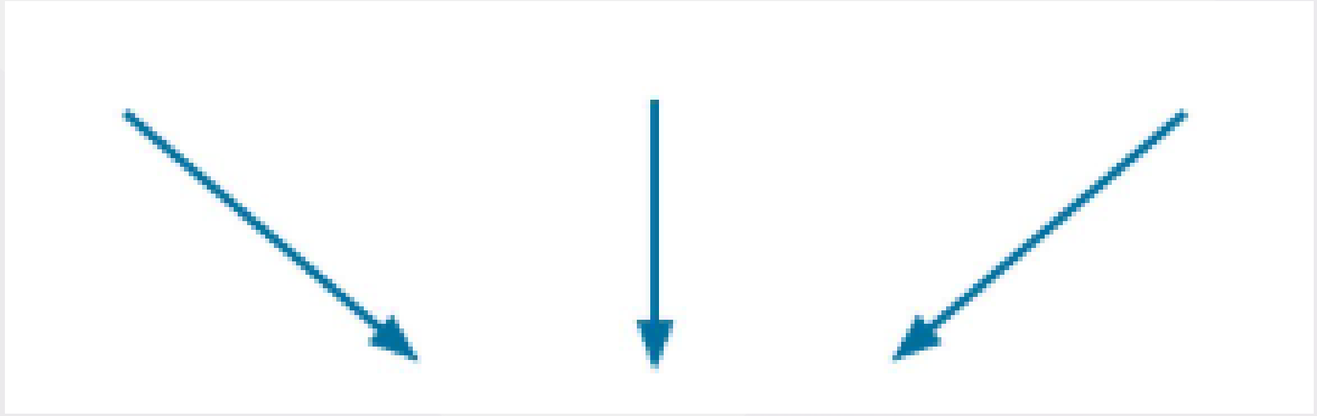
RELATIONSHIP ELEMENTS

- Relationship elements define the location and interconnection of shapes with one another

- Their relational attributes are: direction, position, space, and gravity

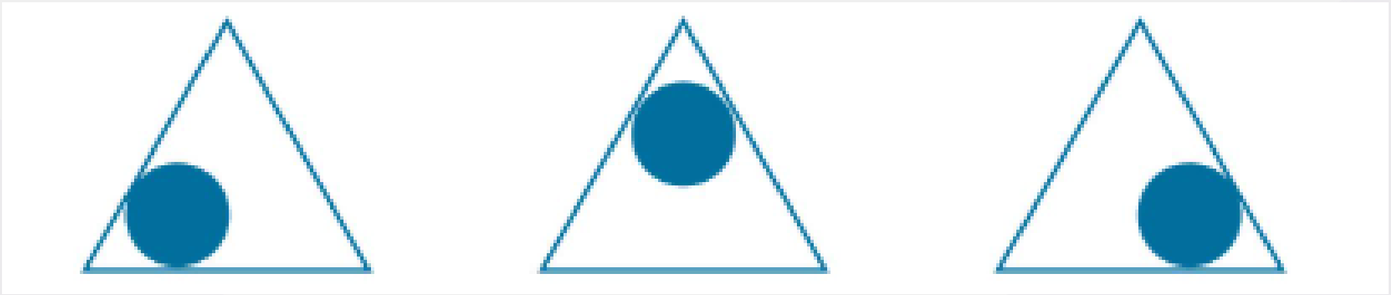
DIRECTION

- Defines the orientation of a shape in relation to other elements in the system
- This includes its orientation with respect to the frame that contains it
- And its orientation relative to the other shapes that make up the site’s design



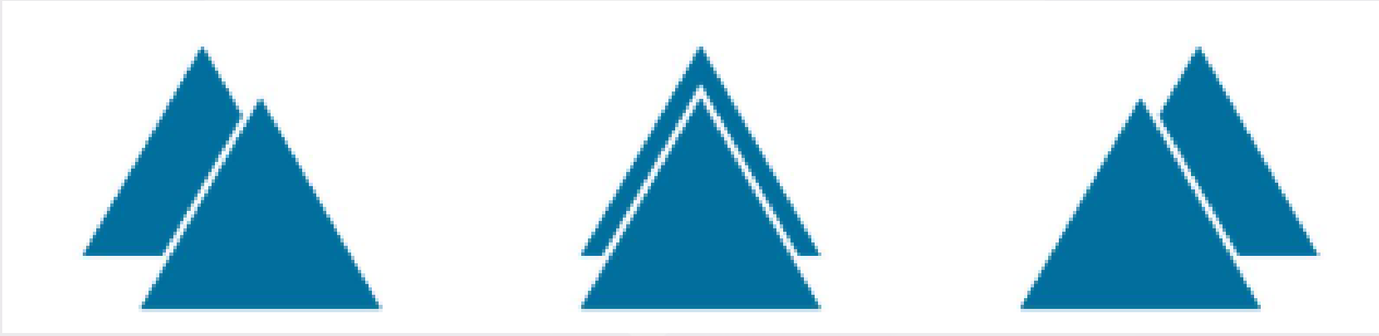
POSITION

- Defines the exact location of a shape
- In relation to the structure in which it is placed



SPACE

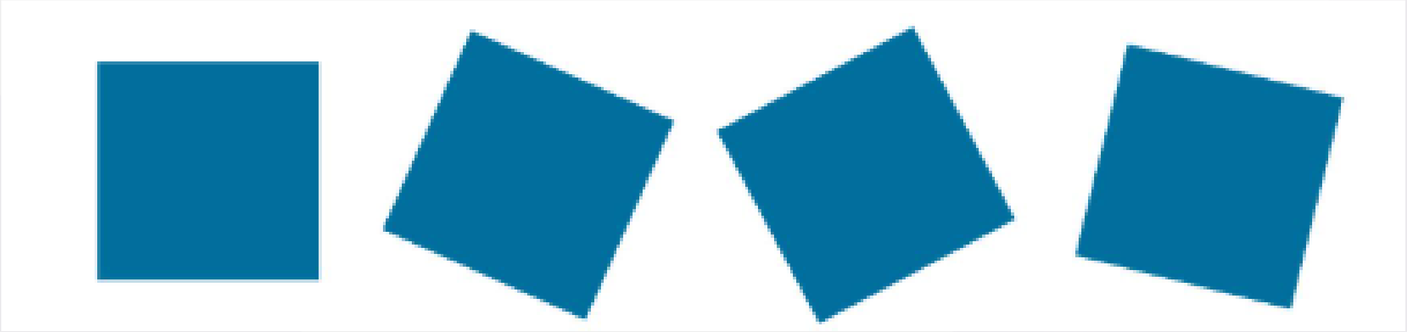
- All shapes occupy space
- Depending on the overall design of the system, this space can create a flat sensation or a sense of depth
- A flat sensation occurs when the occupied area is empty
- A sense of depth appears when the area is filled



GRAVITY

- This property provides a sense of gravity through attributes such as heaviness or lightness, and stability or instability

- These qualities describe how shapes behave visually within the design



PRACTICAL ELEMENTS

- Practical elements focus on the content and scope of a design
- They are interpreted by the mind, allowing us to assign characteristics and meanings to shapes
- In previous slides, the elements used to define each component’s characteristics were listed
- Now we examine the practical elements, whose interpretation is more subjective: representation, meaning, and function

REPRESENTATION

- A shape is representational when it derives from an object in nature or from something created by humans
- Representations may be realistic, closely matching the real-world element they depict
- Or they may appear modified in some of their aspects

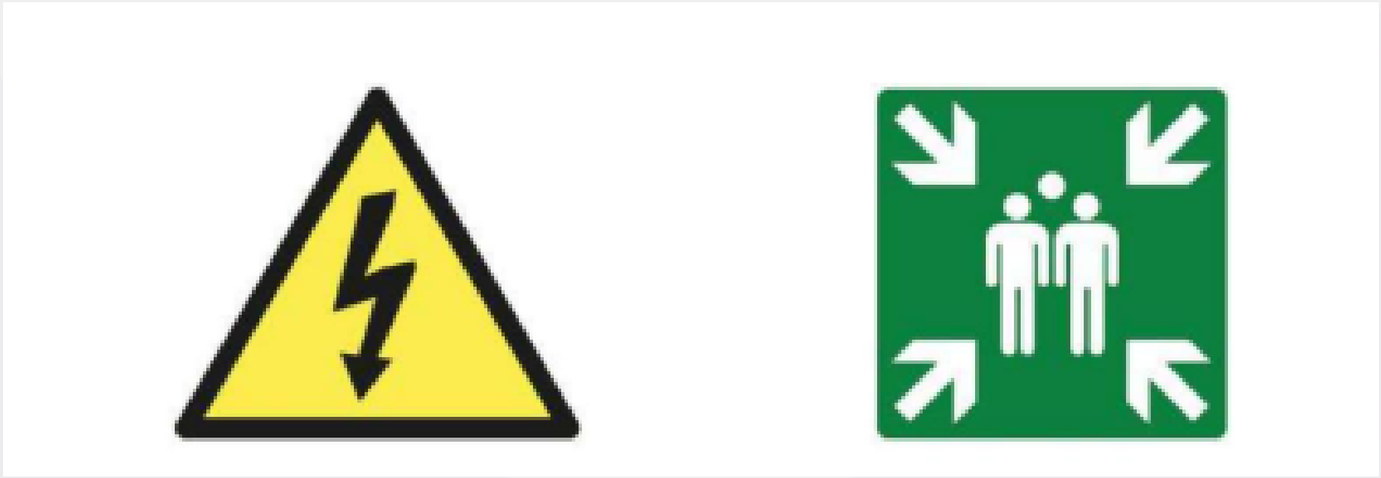
MEANING

- Meaning refers to the interpretation a viewer assigns to a shape
- A shape gains meaning when it evokes an idea, emotion, concept, or association
- This interpretation is subjective, depending on the viewer’s cultural background, experience, and context
- Meaning transforms a simple visual form into something that communicates beyond its appearance

FUNCTION

- Refers to the purpose for which a design is created

- Since every design arises as a solution to a need, its function becomes relevant when the design successfully fulfills the purpose it was intended for



RESOURCES

Unplash (Royalty Free) images: https://unsplash.com/es

Google Fonts: https://fonts.google.com/

ADDITIONAL READINGS

Guidelines in Human Computer Interface (HCI)

Interaction in HCI: Principles, types, and examples

How to create wireframes as a UX Designer

Design Systems 101: What are they and why do they matter?

How to Create a Website Wireframe in Figma (Beginner Tutorial)

How to Wireframe in Figma in 2026

The Psychology of Color in Web Design: What You Need to Know

ACRONIMS

HCI: Human-Computer Interaction
I/O or IO: Input/Output
SME: Small and medium-sized enterprises
UX: User Experience
UI: User Interface

¿ANY QUESTION?